

PROJECT PRESENTATION • PRJ-029

Analytics Q&A Copilot

Natural-language analytics assistant for operational databases, interactive queries, dynamic dashboards, and automated report synthesis.

PRESENTER

PRASANNASURIYA.A.D PSVPEC

REGISTER NO

411723104032

DEPARTMENT & DOMAIN

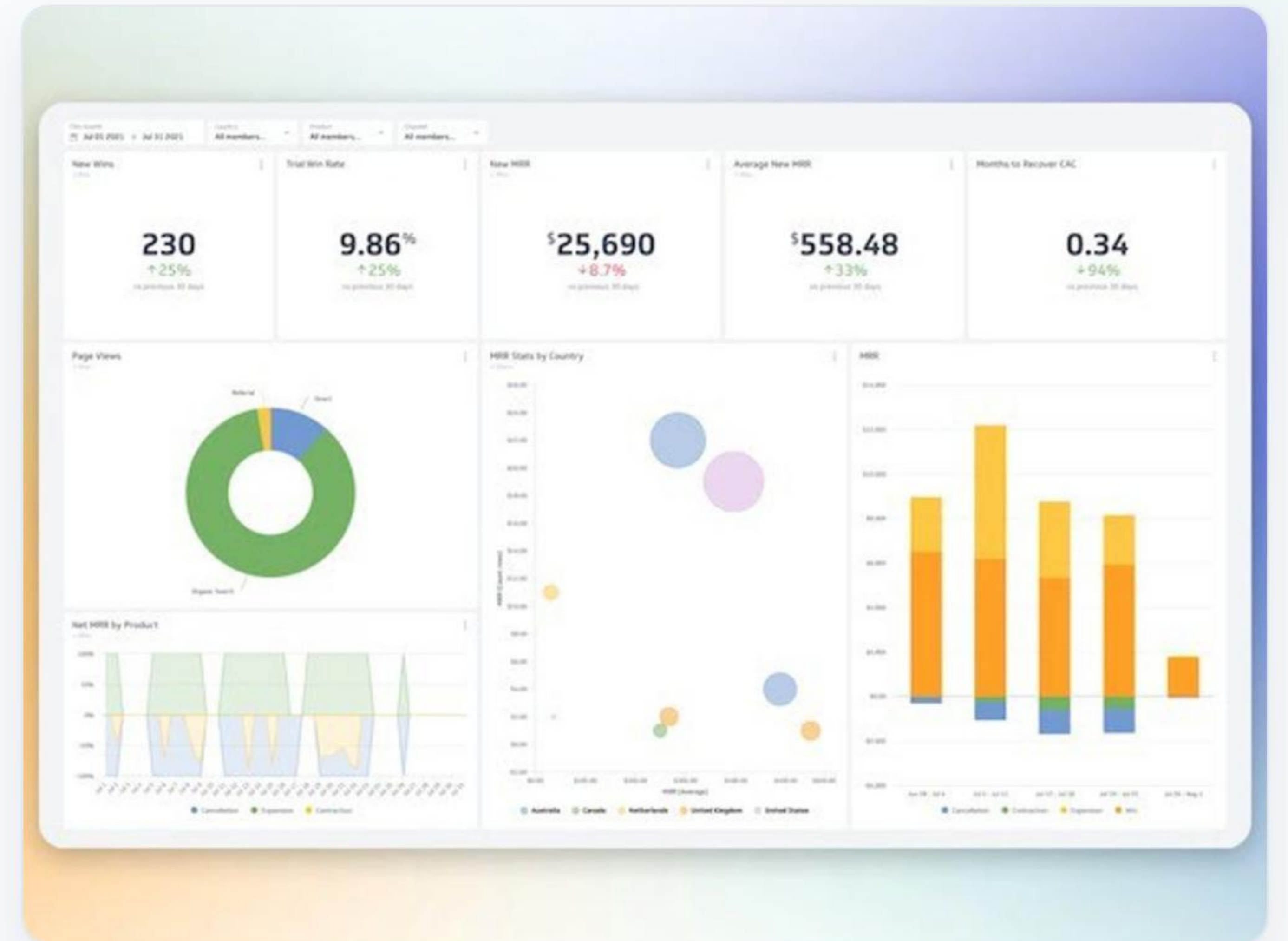
CSE • Business Intelligence & GenAI

PROJECT ID

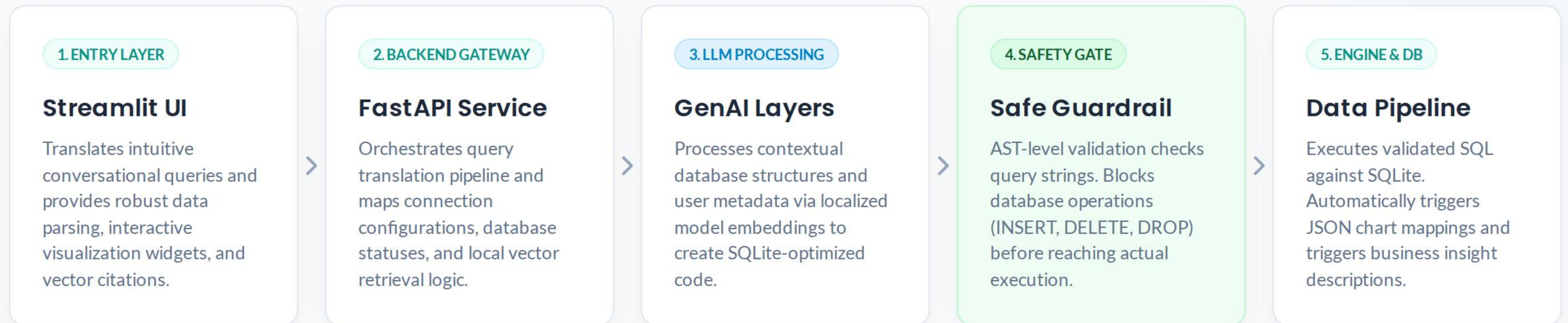
PRJ-029 • RAG Engine

Challenges in Operational Analytics

- ⚠️ **The SQL Knowledge Gap:** Business analysts need rapid insights but lack SQL/relational programming expertise, building operational bottlenecks for IT and backend data teams.
- 🚫 **Limits of Static BI:** Standard dashboard visualizations lack flexibility. They cannot accommodate complex, ad-hoc natural language operations on the fly.
- ⚙️ **GenAI Execution Risks:** Generic open-ended LLMs hallucinate database schemas, creating syntactically broken sql or initiating catastrophic, irreversible write queries (**DROP**, **DELETE**).
- 🎯 **Project Objective:** Architect a lightweight, schema-aware, read-only conversational copilot that parses queries, dynamically builds charts, and describes insights safely.



Core System Architecture



i Modular integration ensures strict decoupling of data ingestion, secure sanitization pipelines, and front-end interface widgets.

| Intelligent Query Generation

-  **Contextual Schema Ingestion:** The dynamic runtime system inspects SQL databases on startup, parsing catalog listings, raw types, indexes, and constraints directly into the prompt payload.
-  **SQLite Constraint Dialects:** Tailored code translation pipeline prevents generic SQL structures. Generates platform-compliant SQLite code specifically (e.g., using `strftime` configurations).
-  **Context-Aware Glossary Mapping:** Connects unstructured requests with precise database properties (e.g., automatically matching expressions like "Active User" to exact SQL column states and metrics).

 Database schema structural mapping and context-aware SQL query construction visual

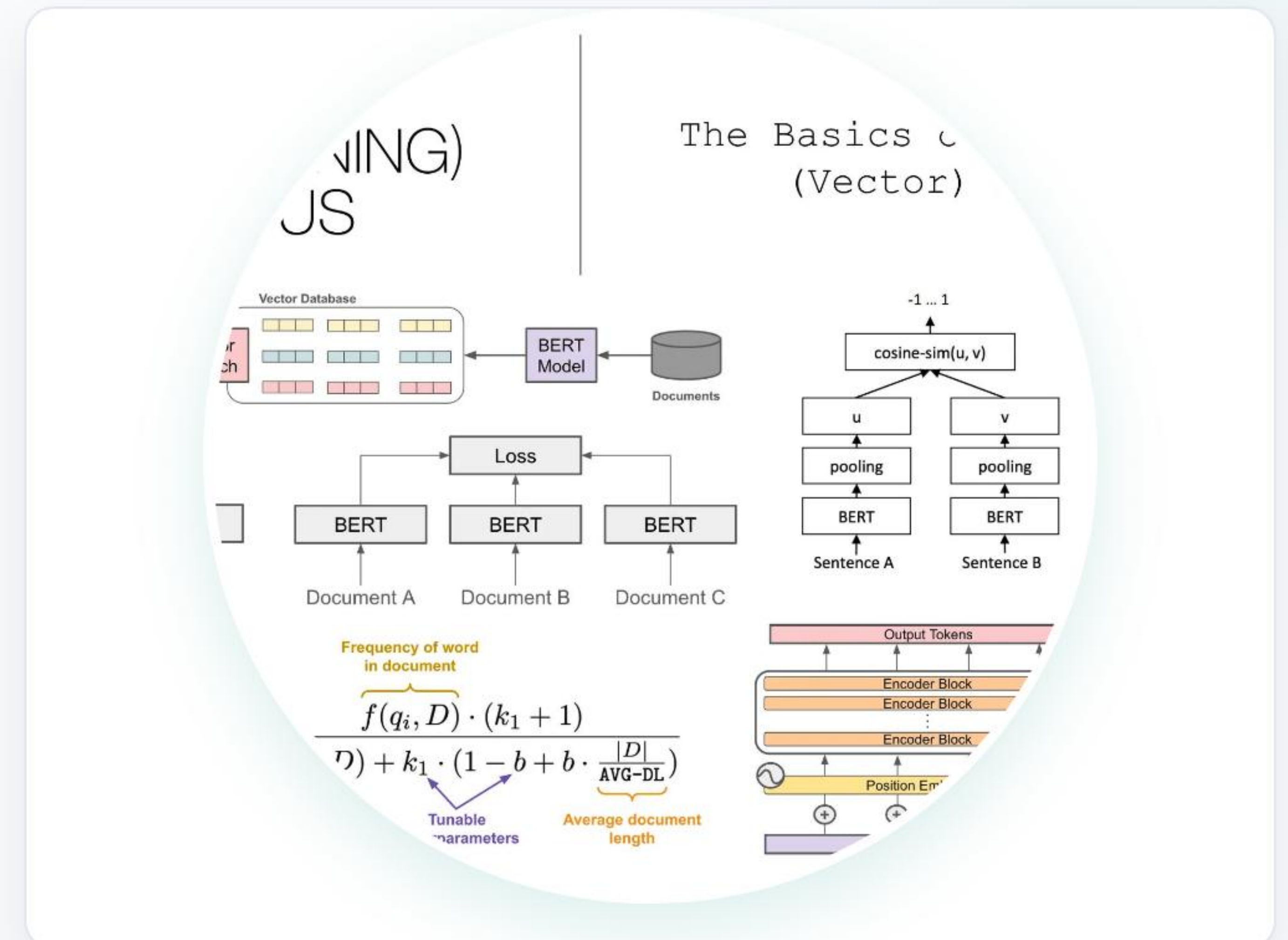
| Robust Execution Security

-  **Read-Only Enforcer Layer:** Rigorous AST interceptors scan compiled SQL payloads, blocking write indicators like **INSERT**, **UPDATE**, **DELETE**, **DROP**, or database manipulation keywords.
-  **Query Syntax Validation:** System checks statement prefixes to guarantee processed scripts only begin with valid SQL read verbs (**SELECT** or **WITH** common table expressions).
-  **SQL Injection Defense:** Neutralizes compound instructions and semicolon separator manipulation to guarantee zero command stacking can be sent down to the DB engine.
-  **Robust Failure Isolation:** Sanitizes engine syntax warnings. Displays clear UI notifications to business operators instead of crashing application systems.

 Cybersecurity shield protecting enterprise database from dangerous queries and injections

Retrieval-Augmented Generation

-  **Dynamic Glossary Indexing:** Seamless document parsing parses glossary terminology, customized calculation variables, and organization glossaries directly.
-  **Text Slicing & FAISS Vectors:** Partitions unstructured definitions using a [RecursiveCharacterSplitter](#), storing vectorized dimensions into a specialized, lightning-fast FAISS storage engine.
-  **Google Embeddings Core:** Leverages the highly responsive Google GenAI Embeddings pipeline ([models/text-embedding-004](#)) for intent mapping.
-  **Citation Auditability:** Automatically lists specific extracted document source snippets in UI modules to help users verify business metrics immediately.

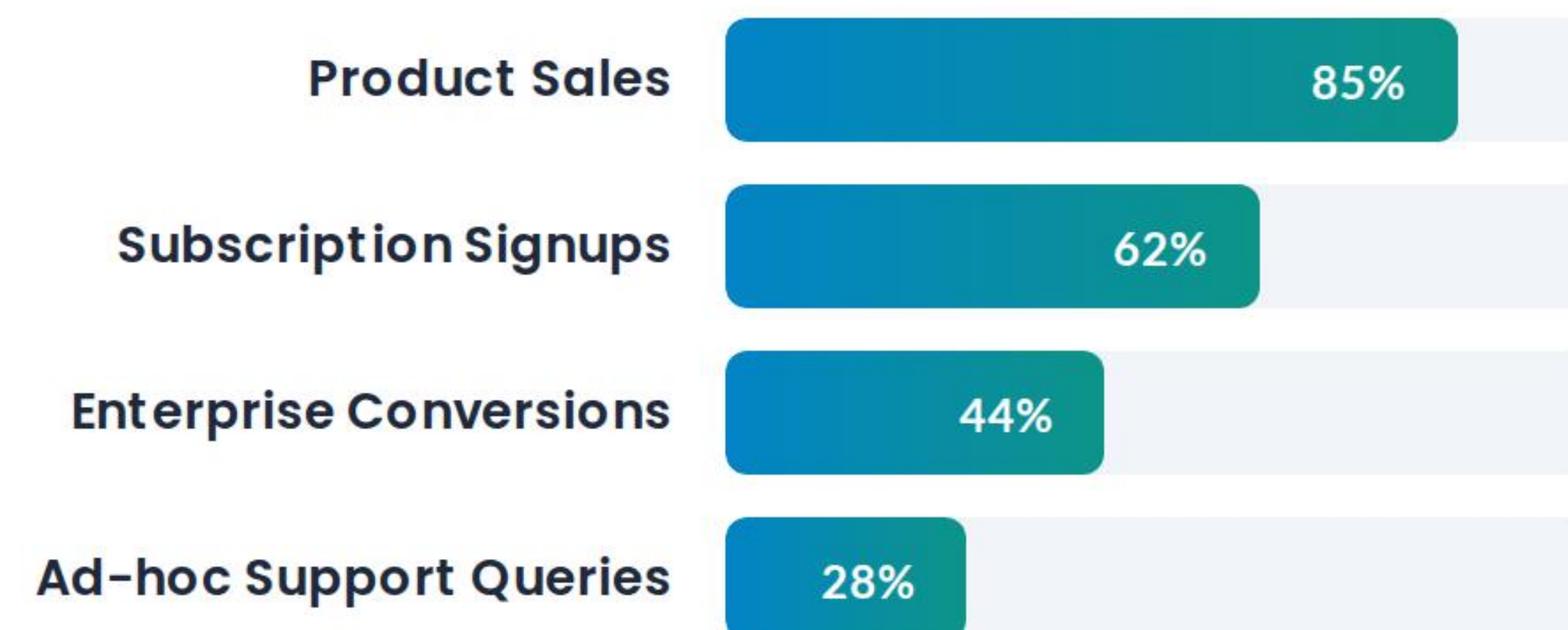


Auto-Charting & Business Summaries

-  **Visual Suggester Engine:** Evaluates dataset structures and shapes dynamically to return JSON payloads classifying target styles (`bar`, `line`, or `pie`).
-  **Interactive Visualization:** Renders dynamically using highly aesthetic, modern, and readable Altair components, complete with interactive tooltips and crisp rounded bars.
-  **Smart Explanations:** Synthesizes raw database outputs into plain English in under 150 words, focusing on totals, key trends, and outliers.

"The model automatically identified 'Revenue' and 'Quarter' categories, selecting a Bar Chart layout and surfacing a 24% month-over-month growth trend."

RENDER OUTPUT EXAMPLE



Technology Stack & Local Deployment



Backend Infrastructure

FastAPI acts as our high-performance data router, exposing robust API paths to handle LLM mapping coordinates and connection pools.



Frontend Interface

Streamlit delivers a modern glassmorphic dashboard interface containing connection statuses, sample questions, and table widgets.



Local Development

Run seamlessly inside standardized environments: FastAPI exposes interfaces on port 8000 while Streamlit serves on port 8501.



Highly Modular Project Structure: `database.py` (parsing) • `vector_store.py` (indexing) • `copilot.py` (security rules & agents).

Automated Verification Results

100%

TEST SUITE PASSED

3 test cases • Execution time: 0.255s

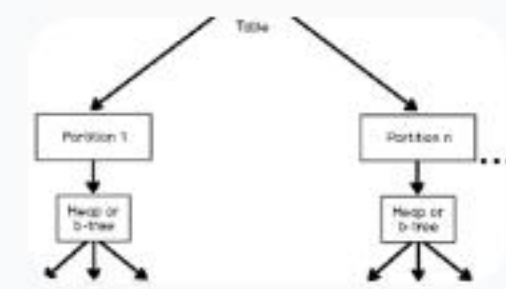
✓ Validation Assertions

Verification Module	Test Purpose	Execution Type	Status
Schema Parser Engine	Confirms accurate table catalog harvesting and type mapping	Unit Test	✓ Passed
SQL Security Interceptor	Asserts write command blockage (INSERT, DROP) and SELECT pass-through	Security Test	✓ Passed
Safe Exec Exception Catching	Validates system traps sqlite syntactical warnings without crashing UI	Robustness Test	✓ Passed

| Image Sources

Thumbnail for https://images.klipfolio.com/website/public/6f1b14b5-1d2d-4a30-ac82-30869ca03ba4/powermetrics-dashboard_0.jpg

www.klipfolio.com
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<https://www.sqlshack.com/wp-content/uploads/2018/03/word-image-25-1.png>

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